

Solution Requirements

Date	26 June 2026
Team ID	SWTID-2026-4638
Project Name	Credit card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Current version does not require user login. Future versions may support secure user authentication for applicants and bank administrators.	Low
2	Authorization Levels	The current system provides a single user role with access to the prediction features. Future versions may include Admin and User roles.	Low
3	External Interfaces	The system integrates with a trained Random Forest machine learning model and preprocessing module to generate credit card approval High predictions.	High
4	Transaction Processing	The system processes applicant information, performs preprocessing, predicts approval status, generates risk audit reports, and stores High prediction history.	High
5	Reporting	The system generates prediction results and detailed risk audit reports that help users understand the approval or denial decision.	Medium

6	Business Rules	Users must provide valid demographic and financial information. The system validates inputs, performs feature engineering, and predicts High eligibility based on the trained model.	High
7	Compliance to Laws or Regulations	The application does not collect confidential credentials. It follows general data privacy principles while securely storing	Medium



S.No	Requirement Category	Requirement Description	Priority
		prediction Medium Regulations history.	
8	Other	The system maintains prediction history and audit logs for future reference, monitoring, and performance analysis.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate credit card approval predictions quickly and provide fast response times.	Average response time < 2 seconds
2	Scalability	The system should support additional users and future cloud deployment.	Supports at least 100 concurrent users in future deployment
3	Security & Data Privacy	Applicant information and prediction records should be securely handled and validated.	No sensitive user data stored; secure API communication
4	Reliability & Availability	The application should operate continuously without major failures.	System availability > 95% during testing
5	Usability & Accessibility	The interface should be intuitive, responsive, and easy to use for all users.	Users can generate suggestions within 3 interaction steps
6	Maintainability & Portability	The code should be modular, reusable, and easy to deploy across different environments.	Modular architecture with reusable services and documented APIs